GO-11 (problem statement): The Exact Two-Consumer Rate–Work Region with Reset Side Information

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Abstract

The complementarity-tax note proved *floors* for the joint description of two consumers whose reset may consult side information; those floors are tight only as $D \to 0$. This document poses the exact-region problem. The coordinate pair itself is not new: per test channel the rate–work gap is $I(\widehat{X}; S)$ (a chain-rule identity that is working folklore in the rate–distortion–equivocation literature since Yamamoto, and appears inside Paper V's achievability proof), and exact scalar-Gaussian regions carrying a conditional-information coordinate with *one* distortion constraint exist in print — most sharply Günlü et al.'s conditional-leakage region under a degradedness ordering, and Wang et al.'s rate–distortion–classification function, which is the scalar instance of our frontier. What our sweeps did not find, and what GO-11 asks for, is the region when (i) the source is a vector read through *two* simultaneous weighted-quadratic consumer distortions, (ii) S is arbitrary jointly Gaussian — neither aligned with the read modes nor degraded relative to any decoder observation — and (iii) the frontier comes with the semidefinite condition under which the complementarity-tax work floor is attained. Even the single-consumer vector endpoint (a vector-Gaussian common-reconstruction function) has no published solution. We state the conjectured structure, the first decidable step, and the verification obligations a registration must meet.

1 The problem

$X \sim \mathcal{N}(0, \Sigma_x)$ on $\mathbb{R}^d$, i.i.d.; consumers A, B with rank-one reads u, v (general PSD P_1, P_2 as the extension); reset side information S, jointly Gaussian with X, available only at erasure, with the model Markov chain $(\widehat{X}_A, \widehat{X}_B) - X - S$ [1]. Define, for $0 < D_A \le \sigma_u^2$, $0 < D_B \le \sigma_v^2$,

$$\mathcal{RW}(D_A, D_B \mid S) = \mathrm{cl} \bigcup_{p(\hat{x}_A, \hat{x}_B \mid x)} \big\{(R, L) : \ R \ge I(X; \widehat{X}_A, \widehat{X}_B), \ L \ge I(X; \widehat{X}_A, \widehat{X}_B \mid S)\big\},$$

the union over joint test channels with $\mathbb{E}\, d_A \le D_A$, $\mathbb{E}\, d_B \le D_B$.

Problem (GO-11). Characterize $\mathcal{RW}(D_A, D_B \mid S)$ exactly — its Pareto frontier, the optimal channel family, and the condition under which the complementarity-tax floors [3] are attained.

What is known (post-sweep accounting). (i) *The one-constraint pair-region is solved, in general.* Paper V's main region theorem characterizes $\{(I(X; \widehat{X}), I(X; \widehat{X} \mid S))\}$ under a single distortion constraint for general alphabets, and its Pareto-channel equation is a Blahut–Arimoto-type fixed point for the weighted objective $\alpha I(X; \widehat{X}) + (1-\alpha) I(X; \widehat{X} \mid S) + \beta\, \mathbb{E} d$ — which, by Fact 1, *is* the scalarization (1) below with $\beta' = 1 - \alpha$. Its solved Gaussian instances: the scalar corner (any jointly Gaussian scalar S) and the mode-aligned vector case (independent modes, per-mode S, one aggregated constraint) with its reset-water-filling frontier [1]. (ii) *The rate-only projection* is the simultaneous-service max-det program ([2]; after the read-plane reduction, the Xiao–Luo / semidefinite-condition literature). (iii) *The floors*: $R \geq \frac{1}{2}\log^{+}(\kappa/(D_A D_B))$, $L \geq \frac{1}{2}\log^{+}(\kappa_S/(D_A D_B))$, floor difference exactly $I(Y;S)$, $Y = B^{\mathsf{T}}X$; rate floor exact iff $\mathrm{diag}(D_A, D_B) \preceq B^{\mathsf{T}}\Sigma_x B$; work floor generally strict at $D > 0$ [3]. (iv) *Operationally*, decode thresholds realize 0.4–0.8$\times$ of the floor-level gaps at finite n on two source families (GO-P-2026-058/059); an exact region would say what the remaining fraction is made of. (v) *Scalar exact regions with one distortion constraint exist in the secrecy literature* — see Section 6 — including the conditional-leakage coordinate itself under degradedness (Günlü et al.) and with the non-degraded scalar Gaussian converse explicitly left open (Villard–Piantanida). GO-11's open content is therefore exactly: general (non-aligned, non-degraded) Gaussian S — already for *one* consumer — and everything two-consumer beyond Paper V's structural union.

2 The organizing identity

Fact 1 (chain rule; folklore)**.** *For any test channel with $(\widehat{X}_A, \widehat{X}_B) - X - S$ Markov, writing $\widehat{X} = (\widehat{X}_A, \widehat{X}_B)$,*

$$I(X;\widehat{X}) - I(X;\widehat{X} \mid S) \;=\; I(\widehat{X};S).$$

Proof. $I(\widehat{X};S) = I(\widehat{X};X,S) - I(\widehat{X};X \mid S) = I(\widehat{X};X) + I(\widehat{X};S \mid X) - I(\widehat{X};X \mid S)$, and $I(\widehat{X};S \mid X) = 0$ by the Markov chain. □

Remark 1 (attribution)**.** *Fact 1 is not ours: it is used verbatim inside Paper V's achievability proof (the binning step) and is working folklore throughout the rate–distortion–equivocation lineage (Yamamoto; Villard–Piantanida's region manipulations). What it* organizes *here is the observation that the region is generated by a single curve of corners $(R, L) = (I,\, I - I(\widehat{X};S))$, so the frontier is the scalarization*

$$\min_{p(\hat{x}|x)} \; I(X;\widehat{X}) - \beta'\, I(\widehat{X};S) \quad \textit{s.t.} \quad \mathbb{E}d_A \leq D_A,\; \mathbb{E}d_B \leq D_B, \qquad \beta' \in [0,1], \tag{1}$$

a distortion-constrained information bottleneck with the reset context S as relevance — equivalently, in equivocation coordinates, a maximize-the-leakage member of the Yamamoto family with the orientation that helps *the S-holding party. $\beta' = 0$ recovers the simultaneous-service rate program; $\beta' = 1$ the work-only endpoint.*

Remark 2 (convexity boundary — a warning)**.** *For $\beta' \in [0,1]$,* (1) *is the convex combination $\alpha I(X;\widehat{X}) + (1-\alpha) I(X;\widehat{X} \mid S)$ of the two coordinates ($\alpha = 1 - \beta'$), each convex in the channel (the conditional coordinate by the per-s decomposition verified in VI-8), so the scalarization stays convex and traces the entire frontier. The difference form $I(X;\widehat{X}) - \beta' I(\widehat{X};S)$ is* not *manifestly convex on its own ($I(\widehat{X};S)$ is convex in the channel, not concave); any argument pushing $\beta' > 1$ leaves the convex regime and must be justified separately.*

3 Conjectures and questions

The sweeps (Section 6) make the following *expected by analogy* rather than surprising: every ingredient exists separately in print; no assembled solution does.

Conjecture 1 (Gaussian sufficiency). *The frontier of* (1) *is attained by jointly Gaussian channels* $\widehat{X} = KX + W$*; consequently the region is the closure of* $\{(R(\Sigma), R(\Sigma) - I_\Sigma(\widehat{X}; S))\}$ *over error covariances* $0 \preceq \Sigma \preceq \Sigma_x$ *meeting the trace constraints, with both coordinates explicit determinant functionals of* Σ. Expected route: extend the conditional-EPI arguments that prove Gaussian optimality in the degraded scalar cases (Günlü et al.; Sankar et al.'s simultaneous rate-and-leakage optimality) and the linear-policy optimality machinery of the mixed-quadratic Gaussian literature; the open difficulty is precisely the non-degraded, non-aligned coupling, where Villard–Piantanida's scalar converse is already open.

Conjecture 2 (Two-consumer reset water-filling — possibly optimistic). *In the* S*-whitened read frame, the frontier is traced by an* α*-parametrized allocation program generalizing Paper V's reset water-filling: the* $\alpha = 1$ *endpoint re-derives the max-det rate optimum, the* $\alpha = 0$ *endpoint tilts distortion toward the directions of the read plane that* S *resolves least (the operational allocation signature of GO-P-2026-058/059), and for scalar-pair instances the program reduces to a two-mode quadratic of the Paper V vector-frontier type.* Caution: with Σ_x, $\Sigma_{X|S}$, and the read operators not simultaneously diagonalizable, three mutually non-commuting matrices compete; the analogous regime of the 2026 rate–distortion–deception work found no closed form. A clean water-filling may exist only under an alignment condition, with the general case a semidefinite frontier.

Conjecture 3 (Exactness condition for the work floor). *The work floor* $\frac{1}{2}\log(\kappa_S/(D_A D_B))$ *is attained iff a semidefinite condition of the form* $\operatorname{diag}(D_A, D_B) \preceq B^\mathsf{T}\Sigma_{X|S}B$ *holds* and *the optimal channel decouples from* S*'s residual — expected to fail at every* $D > 0$ *when* S *has an* X*-independent component (*$\tau^2 > 0$*; matching the strictness remark of [3] and the scalar corner's* $D > 0$ *gap), with equality exactly in the* $D \to 0$ *limit. The qualifier is forced by Proposition 1: an* X*-measurable* S *is maximally informative yet the single-consumer conditional floor is attained there at every* $D \le \frac{1}{2}$ *— the strictness belongs to the encoder's ignorance of* S*'s residual, not to informativeness.*

Question 1 (Single-consumer vector case; endpoint = a vector-Gaussian common-reconstruction function). *Even* $m = 1$ *is open: for one rank-one read of a* d*-dimensional source with* arbitrary *jointly Gaussian* S*, what is the frontier — and already its work-only endpoint* $L_A(D \mid S) = \min I(X; \widehat{X} \mid S)$*, which equals Steinberg's common-reconstruction rate–distortion function? The targeted sweep found* no published vector-Gaussian CR function under any formulation *— not even unweighted distortion, not even the mode-aligned case solved in Paper V. This is Paper V's declared "general Gaussian coupling" open item, specialized to rank-one reads.*

The first decidable step (rank-one $P = ww^\mathsf{T}$) is now settled *— see Proposition 1: marginalization is strictly suboptimal, the rank-one case does* not *collapse to the scalar corner, and the mechanism (the encoder's implicit knowledge of the* X*-measurable part of* S*) reshapes what the general answer must look like.*

4 The marginalization dichotomy, settled

Throughout this section $m = 1$: one rank-one read $Y = w^\mathsf{T}X$, distortion $\mathbb{E}(Y - \widehat{Y})^2 \le D$. Two reductions first. *(WLOG scalar reproduction.)* Replacing $\widehat{X}$ by $\widehat{Y} := w^\mathsf{T}\widehat{X}$ leaves the distortion unchanged and can only decrease $I(X; \cdot \mid S)$ (conditional data processing), and $\widehat{Y} - X - S$ persists; so reproductions are scalar. *(Marginalized class = scalar corner.)* If the channel depends on X only through Y (i.e., $\widehat{Y} - Y - (X, S)$), then given (Y, S), $\widehat{Y}$ is independent of X, so $I(X; \widehat{Y} \mid S) =$

$I(Y;\widehat{Y} \mid S)$, and the class optimum is exactly the scalar corner value of [1] for the pair (Y,S) — the corner's converse applies to every $p(\hat{y} \mid y)$, discrete or not.

Proposition 1 (Marginalization is strictly suboptimal)**.** *Let $d=2$, $X \sim \mathcal{N}(0,I_2)$, $Y=X_1$, and $S=X_1+X_2$ (so $\rho_{YS}^2=\frac{1}{2}$ and S is X-measurable). For $0<D<1$:*

(i) the marginalized-class optimum is the scalar corner $L_{\mathrm{marg}}(D)=\frac{1}{2}\log_2\frac{1+D}{2D}$;

(ii) the vector problem has the exact value

$$L_{\mathrm{vec}}(D) \;=\; \tfrac{1}{2}\log_2^{+}\frac{1}{2D} \;=\; R_{Y|S}(D),$$

Gray's conditional rate–distortion function of (Y,S) ($\mathrm{Var}(Y \mid S)=\frac{1}{2}$), attained by the linear channel $\widehat{Y}=(1-D)X_1+DX_2+N$, $N\sim\mathcal{N}(0,D(1-2D))$, for $D\le\frac{1}{2}$, and by $\widehat{Y}=S/2$ (zero conditional content) for $D\ge\frac{1}{2}$;

(iii) hence the strict gap $L_{\mathrm{marg}}(D)-L_{\mathrm{vec}}(D)=\frac{1}{2}\log_2(1+D)>0$ on $(0,\frac{1}{2}]$, and equal to the full marginalized value on $[\frac{1}{2},1)$ (where $L_{\mathrm{vec}}=0$; the absolute gap is decreasing there, but it is all of L_{marg}).

Proof. (i) is the reduction above with Paper V's scalar-corner formula at $\sigma^2=1$, $\rho^2=\frac{1}{2}$.

(ii) *Converse.* For any admissible $\widehat{Y}$ (Markov $\widehat{Y}-X-S$), $I(X;\widehat{Y}\mid S)\ge I(Y;\widehat{Y}\mid S)$, and the induced conditional channel $p(\hat{y}\mid y,s)$ is feasible for Gray's conditional problem at the same distortion, so $I(Y;\widehat{Y}\mid S)\ge R_{Y|S}(D)=\frac{1}{2}\log_2^{+}\big(\mathrm{Var}(Y\mid S)/D\big)=\frac{1}{2}\log_2^{+}(1/(2D))$. *Achievability.* For $D\le\frac{1}{2}$ take $\widehat{Y}=(1-D)X_1+DX_2+N$: distortion $D^2+D^2+D(1-2D)=D$; $\mathrm{Var}(\widehat{Y})=1-D$, $\mathrm{Cov}(\widehat{Y},S)=1$, $\mathrm{Var}(\widehat{Y}\mid S)=1-D-\frac{1}{2}=\frac{1}{2}-D$, so $I(X;\widehat{Y}\mid S)=\frac{1}{2}\log_2\frac{(1-2D)/2}{D(1-2D)}=\frac{1}{2}\log_2\frac{1}{2D}$. For $D\ge\frac{1}{2}$, $\widehat{Y}=S/2$ is a deterministic function of X, has distortion $\mathbb{E}\big((X_1-X_2)/2\big)^2=\frac{1}{2}\le D$, and is constant given S, so $I(X;\widehat{Y}\mid S)=0$.

(iii) Subtract. Numerical falsification (pre-assertion, logged): the closed forms reproduce under direct minimization over all linear channels (40-start SLSQP, five D values, agreement 2×10^{-4}), and a discretized conditional Blahut–Arimoto optimizer over *unrestricted* channels matches L_{vec} within grid error (0.372 vs 0.368 at $D=0.3$; 0.005 vs 0 at $D=0.5$) while the Y-only optimizer matches the corner (0.560 vs 0.558; 0.293 vs 0.293). □

Remark 3 (The mechanism: writing in ink the eraser can read)**.** *When S is X-measurable, the encoder implicitly* knows *S, and the common-reconstruction Markov constraint costs nothing beyond Gray's conditional problem: the optimal channel deliberately correlates the record with S — at $D\ge\frac{1}{2}$ the record* is *a function of the reset context, erasure-free by construction. Steinberg's scalar strictness (CR $>$ conditional RD at $D>0$) is thus not intrinsic to the CR constraint; it is the price of an encoder that cannot see S's non-read directions. General jointly Gaussian $S=\gamma^{\mathsf{T}}X+U$ ($\mathrm{Var}\,U=\tau^2>0$) interpolates: the encoder commands the X-part $\gamma^{\mathsf{T}}X$ but not U, suggesting the refined question below. Operationally this is a new prediction for any GO-11 face: optimal joint records should tilt* toward *the reset context's X-measurable directions — the encoder-side complement of the eraser-side allocation tilt measured in GO-P-2026-058/059.*

Question 2 (Refined form of Question 1 — settled in Section 5)**.** *For $S=\gamma^{\mathsf{T}}X+U$, is the vector CR function exactly a Kaspi-type encoder-side-information object driven by the encoder-accessible* *statistic $\gamma^\mathsf{T} X$ — reducing to Gray at $\tau^2 = 0$ (Proposition 1) and to Steinberg's scalar formula when $\gamma \parallel w$?* Answer (Theorem 1–Corollary 2): structurally yes — only the pair $(w^\mathsf{T} X, \gamma^\mathsf{T} X)$ matters, in any dimension; formula-wise no — the value is a new closed-form interpolating family (2), provably not a functional of the (Y, S) margin.

5 The refined question, settled: pair sufficiency and the cubic family

Normalize $\mathrm{Var}(w^\mathsf{T} X) = 1$ (rescaling D), write $Y = w^\mathsf{T} X$, $V = \gamma^\mathsf{T} X/\sigma_\gamma$ with $\rho := \mathbb{E}[YV]$, $|\rho| < 1$, and rescale S (invariantly for all mutual informations) to $S = V + U$, $U \sim \mathcal{N}(0, \tau^2)$, $\tau^2 > 0$, with X, U, and all channel randomness (including any resampling randomness below) *mutually* independent — the joint form is what the S-kernel argument uses.

Theorem 1 (Pair sufficiency)**.** *In the single-consumer problem $L(D) = \min\{I(X; \widehat{Y} \mid S) : \widehat{Y} - X - S,\ \mathbb{E}(Y - \widehat{Y})^2 \le D\}$, it is without loss of optimality to let the channel depend on X only through $T = (Y, V)$, and then $I(X; \widehat{Y} \mid S) = I(T; \widehat{Y} \mid S)$. Consequently $L(D)$ depends on $(\Sigma_x, w, \gamma, \tau^2)$, in any ambient dimension d, only through (ρ, τ^2).*

Proof. Given an admissible $p(\hat{y} \mid x)$, draw $\widehat{Y}'$ from the conditional law $p(\hat{y} \mid T = t)$ using fresh randomness. (a) $(T, \widehat{Y}') \overset{d}{=} (T, \widehat{Y})$, so the distortion (a functional of $(Y, \widehat{Y})$) is unchanged. (b) $S = V + U$ with U independent of $(X, \widehat{Y})$ and of $(X, \widehat{Y}')$, so in both joints $p(s \mid t, \hat{y}) = p_U(s - v)$: hence $(T, \widehat{Y}', S) \overset{d}{=} (T, \widehat{Y}, S)$. (c) Given T, $\widehat{Y}'$ is independent of (X, S), so $I(X; \widehat{Y}' \mid S) = I(T; \widehat{Y}' \mid S) + I(X; \widehat{Y}' \mid T, S) = I(T; \widehat{Y} \mid S) \le I(X; \widehat{Y} \mid S)$, the last step because T is a function of X. (d) $\widehat{Y}' - X - S$ holds since $\widehat{Y}'$ is a function of $(X, \text{fresh noise})$ with the noise independent of (X, U). The same computation applied to a T-channel directly gives $I(X; \widehat{Y} \mid S) = I(T; \widehat{Y} \mid S)$. □

Theorem 2 (The exact single-consumer CR function, rank-one read)**.** *Write $s = 1 + \tau^2$. For $0 < D < 1$,*

$$L(D) = \tfrac{1}{2}\log_2 g^\star, \qquad g^\star = \frac{(D + s - \rho^2) + \sqrt{(D + s - \rho^2)^2 - 4Ds(1 - \rho^2)}}{2Ds}, \tag{2}$$

the larger root of $P(g) = Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2)$; since $P(1) = (D - 1)\tau^2 < 0$, exactly one root exceeds 1, so $g^\star$ is well defined and unique. The optimum is attained by the jointly Gaussian channel $\widehat{Y} = aY + bV + N$, $a = \dfrac{g^\star - 1}{g^\star}$, $b = \dfrac{(g^\star - 1)\rho}{g^\star k}$, $k = g^\star(1+\tau^2) - 1$, $N \sim \mathcal{N}\big(0,\ Q/(g^\star - 1)\big)$, $Q = a^2 + b^2 + 2ab\rho - (a\rho + b)^2/(1+\tau^2)$. Anchors, algebraically: P factors as $(Dg - 1)(sg - 1)$ at $\rho = 0$ (classical, $g^\star = 1/D$); as $(g - 1)(Dg - (1 - \rho^2))$ at $\tau^2 \to 0$ (Gray, matching Proposition 1, including the achieving channel); and gives $g^\star = (D+\tau^2)/(D(1+\tau^2))$ at $\rho^2 \to 1$ (Steinberg's corner at $\rho_{YS}^2 = 1/(1+\tau^2)$).

Proof (complete; converse closed by the verification pass). Achievability. For the stated channel, $I(T; \widehat{Y} \mid S) = \frac{1}{2}\log_2\big(\mathrm{Var}(\widehat{Y} \mid S)/\operatorname{Var} N\big)$ (jointly Gaussian; $S \perp N$ given T), and $P(g^\star) = 0$ is the statement that the distortion equals D while $\mathrm{Var}(\widehat{Y} \mid S)/\operatorname{Var} N = g^\star$.

Converse. By Theorem 1 restrict to T-channels. For any admissible joint, with $c = \mathbb{E}[Y\widehat{Y}]$, $q = \mathbb{E}[V\widehat{Y}]$, $v = \operatorname{Var}\widehat{Y}$ (WLOG $\mathbb{E}\widehat{Y} = 0$): mutual independence of $(X, U, \text{channel noise})$ forces the moment identity $\mathbb{E}[S\widehat{Y}] = \mathbb{E}[V\widehat{Y}] = q$ (Gaussian Markovity at the second-moment level); the distortion constraint reads $1-2c+v \le D$; and $L \ge B(c,q,v) := \frac{1}{2}\log_2\big(\det \Sigma_{T|S}/\det \Sigma_e(c,q,v)\big)$, with Σ_e the error covariance of the best *linear* estimate of T from $(\widehat{Y}, S)$ — valid for arbitrary reproduction alphabets by Gaussian entropy maximization. The minimization of B closes exactly: in the variables $(a,b) = \Sigma_T^{-1}(c,q)^\mathsf{T}$, $n = v-(a^2+b^2+2ab\rho)$, the determinant identity

$$\frac{\det \Sigma_{T|S}}{\det \Sigma_e} = \frac{\operatorname{Var}(aY+bV \mid S)+n}{n}$$

holds (the S-column of the moment matrix equals the V-column plus $\tau^2 e_S$), so the moment program *is* the linear-channel program; $B \to \infty$ on the PSD boundary $n \to 0$ and at the distortion boundary, the first-order system is linear in (a,b) at fixed g with the unique solution above, and $P(1) < 0$ pins the unique stationary $g^\star > 1$. Hence $\min B = \frac{1}{2}\log_2 g^\star$, matched by achievability. [Derivation completed analytically by the fresh-context verification pass; record and instance-level numerics (agreement to 3×10^{-9} bits at eight fresh instances; 25,500-point root-uniqueness sweep) in the NOVELTY addendum.] □

Remark 4 (Scope — what is and is not settled)**.** *Theorem 2 answers Question 1's* work-only endpoint *for rank-one reads in full generality of $(d, \Sigma_x, \gamma, \tau^2)$: by Theorem 1 the ambient problem is exhausted by (ρ, τ^2), and Gaussian channels suffice* at that endpoint*. It does* not *settle Conjectures 1–2: the full (R, L) frontier for $m = 1$ is a nontrivial curve — the L-optimal channel pays strict excess rate over $R_{\min} = \frac{1}{2}\log_2(1/D)$ (e.g.* 0.040 *bits at $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$; verification pass), so* no single channel attains both corners*. This is itself a finding: unlike Paper V's scalar corner, the rank-one-read problem with non-aligned side information has a real rate–work tradeoff already at $m = 1$. The moment method visibly extends to the weighted objective $\alpha R + (1-\alpha)L$ (both max-entropy bounds hold with the same three moments) — tracing that frontier is the named next step, before the $m \ge 2$ case, for which the pair-sufficiency argument generalizes verbatim ("(P-read components, $\gamma^\mathsf{T} X$) suffice"), collapsing two rank-one consumers to at most three coordinates.*

The $m = 1$ frontier

Theorem 3 (The exact rate–work region, $m = 1$, rank-one read)**.** *For $\rho^2 \in (0,1)$, $\tau^2 > 0$, $0 < D < 1$, the region $\{(R, L)\}$ of Section 1 restricted to one consumer is the closed convex set whose Pareto frontier is traced by $\alpha \in [0,1]$ as follows. Let $(\gamma_0, \gamma_1) \in (0,1]^2$ solve, together with (a, b, n), the stationarity system*

$$a = 1 - \alpha\gamma_0 - (1-\alpha)\gamma_1, \qquad m = \frac{a\rho}{s-(1-\alpha)\gamma_1}, \qquad b = (1-\alpha)\gamma_1 m,$$

$$n = D - h(a,b), \qquad \gamma_0 = \frac{n}{Q_0 + n}, \qquad \gamma_1 = \frac{n}{Q_1+n},$$

where $h(a,b) = (1-a)^2 - 2(1-a)b\rho + b^2$, $Q_0 = a^2+b^2+2ab\rho$, $Q_1 = Q_0 - (a\rho+b)^2/s$, and $m := (a\rho+b)/s$ (solution attained by the weighted optimum and unique at every α*: $P(1) < 0$ at $\alpha = 0$, Proposition 2 for $\alpha \in (0,1]$ — so the trace of the Pareto boundary is unconditionally complete). The frontier point is*

$$\big(R(\alpha), L(\alpha)\big) = \Big(\tfrac{1}{2}\log_2 \tfrac{1}{\gamma_0},\ \tfrac{1}{2}\log_2 \tfrac{1}{\gamma_1}\Big),$$

achieved by $\widehat{Y} = aY + bV + N$, $N \sim \mathcal{N}(0,n)$. Endpoints: $\alpha = 1$ recovers the classical reverse channel ($b = 0$, $a = 1-D$, $R = \frac{1}{2}\log_2 \frac{1}{D}$); $\alpha = 0$ recovers Theorem 2's channel and value.

Proof (converse mechanism as in Theorem 2). Both coordinates are convex in the channel (the conditional coordinate by the per-s decomposition, VI-8), so the scalarization $\alpha R+(1-\alpha)L$ traces the full Pareto boundary. For any admissible channel with moments (c, q, v), both max-entropy bounds hold on the same three moments: $R \geq \frac{1}{2}\log_2(\det\Sigma_T/\det\Sigma_{e0})$ (estimation from $\widehat{Y}$ alone) and $L \geq \frac{1}{2}\log_2(\det\Sigma_{T|S}/\det\Sigma_{e1})$ (from $(\widehat{Y}, S)$, with the Markov moment identity $\mathbb{E}[S\widehat{Y}] = q$). The two determinant identities

$$\frac{\det\Sigma_T}{\det\Sigma_{e0}} = \frac{Q_0+n}{n}, \qquad \frac{\det\Sigma_{T|S}}{\det\Sigma_{e1}} = \frac{Q_1+n}{n}$$

(the first by the same Schur computation as the second) make the weighted moment program identical to the weighted linear-channel program; it is coercive on its boundaries, and its first-order system — using $Q_{0,a}-h_a = 2$, $Q_{0,b}-h_b = 2\rho$, $Q_{1,a}-h_a = 2(1-\rho m)$, $Q_{1,b}-h_b = 2(\rho-m)$ — reduces linearly to the displayed relations. Achievability is the stated Gaussian channel, which meets both bounds with equality. [Numerics logged pre-assertion, seed 20260806: the system holds at the multi-start optimum and the moment program agrees to 10^{-6} bits at 20 $(\rho^2, \tau^2, D, \alpha)$ combinations; the α-sweep is monotone between the corners.] $\square$

Corollary 1 (Misalignment always opens a tradeoff). *For $\rho^2 \in (0,1)$ and $\tau^2 > 0$ the frontier is nondegenerate. The L-optimal moments are unique with $b \neq 0$ (Theorem 2; $b = (g^\star-1)\rho/(g^\star k) \neq 0$), and the R-optimal moments are unique with $b = 0$ (at $\alpha = 1$ the elimination gives $b(1-\rho^2) = 0$, and then $(a, n) = (1-D,\ D(1-D))$ is forced). Hence any channel attaining $L_{\min}$ carries the $b \neq 0$ moments, so its rate is at least B_0 evaluated there, which strictly exceeds $R_{\min}$ (B_0's argmin being unique with $b = 0$) — and symmetrically. Both endpoint excesses are strictly positive (at $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$: $R(0) - R_{\min} = 0.0400$, $L(1) - L_{\min} = 0.0349$ bits). The single-corner collapses within the parameter space are exactly the alignment degeneracies: $\rho^2 \to 1$ (Paper V's scalar corner) and $\rho = 0$ (the frontier collapses to the single corner $L_{\min} = R_{\min}$; $\tau^2 \to \infty$ is the same collapse in the limit). Closed-form endpoint values: $L(1) = \frac{1}{2}\log_2\big[((1-D)(1-\rho^2/s)+D)/D\big]$ and $R(0)$ from Theorem 2's channel. The same strictness holds in the $\tau^2 = 0$ instance computations of Proposition 1 for $D < \frac{1}{2}$ (e.g. $R(0) = \frac{1}{2}\log_2[(1-D)/(D(1-2D))] > \frac{1}{2}\log_2(1/D)$), so even a fully X-measurable context trades rate against work once misaligned.*

This settles Conjecture 1 affirmatively for $m = 1$, rank-one reads (Gaussian sufficiency for the whole frontier), and replaces Conjecture 2's $m = 1$ shadow with the explicit two-water-level system; the $m \geq 2$ statements remain open.

Corollary 2 (Answer to the refined question). *(i)* Structurally yes*: the value is driven entirely by the encoder-accessible statistic $\gamma^\mathsf{T} X$ (pair sufficiency), reducing to Gray at $\tau^2 = 0$ and to Steinberg at $\gamma \parallel w$. (ii)* Formula-wise no*: L is* not *a functional of the joint law of (Y, S) — the instances $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$ and $(\frac{3}{4}, \frac{1}{2})$ have identical (Y, S) margins ($\rho^2_{YS} = \frac{1}{2}$) yet $L = 1.1610$ vs 1.2105 bits at $D = 0.1$ (0.3685 vs 0.5228 at $D = 0.3$), both strictly below the Steinberg value of the margin (1.2297; 0.5577). In particular the vector CR function is not Steinberg's formula composed with any modified pair, nor any known scalar object: it is the new interpolating family* (2)*, monotone (observed) in the encoder-access split τ^2 at fixed margin.*

Question 3 (Operational face). *Do decode thresholds track the exact frontier (not just the floors) — in particular, does the measured* 0.4–0.8× *realized fraction of GO-P-2026-058/059 decompose into (exact-region gap) + (finite-n instrument loss), with the first term predicted by Conjecture 2?*

6 Delineation (sweeps of 2026-08-04, record in `goll-conditional-region-NO`

In-repo (Paper V). The one-constraint pair-region is Paper V's, completely: main region theorem (general alphabets), Pareto-channel fixed point (= our scalarization at $\beta' = 1 - \alpha$), scalar corner, and mode-aligned reset water-filling. Paper V's multi-consumer theorem gives only the structural union for $m \geq 2$; its "general Gaussian coupling" limitation names Question 1's territory as open.

IB / task-oriented compression. The unconstrained Gaussian tradeoff of (1) is the solved Gaussian information bottleneck [4]. The *scalar single-distortion instance of our frontier is prior art*: Wang et al.'s rate–distortion–classification function [10] minimizes $I(X;\widehat{X})$ under an MSE cap and an $I(\widehat{X};S)$ floor (their $H(S \mid \widehat{X}) \leq C$), with our Markov chain, solved in closed scalar/bivariate-Gaussian form — including the one-constraint-active-at-a-time structure that matches the scalar corner. The strongest machinery match is Liu et al.'s semantic rate–distortion function [11]: two *simultaneous* quadratic constraints on a Gaussian source, solved as an error-covariance convex program with reverse water-filling — but its second coordinate is an MSE on the state, not an information quantity. The sign-flipped (minimize-leakage) family has scalar Gaussian closed forms since Rebollo-Monedero et al. (2010) and Sankar et al. [8], and a special-case vector water-filling tilting distortion toward the modes that leak least [12] — the mirror of Conjecture 2's allocation.

Secrecy / equivocation (the same family, same orientation). Maximizing source equivocation at an S-holding party is minimizing our L up to $H(X \mid S)$, so the Yamamoto lineage [5] is not a cousin but the family itself. Exact Gaussian regions exist and are all scalar with one distortion constraint: Villard–Piantanida [6] (exact only when the eavesdropper has no side information; the non-degraded scalar Gaussian converse *explicitly left open*); Schieler–Cuff (Gaussian solved only without adversary side information); and, sharpest, Günlü–Schaefer–Boche–Poor [7], whose privacy-leakage coordinate $I(X;W \mid S)$ *is* our conditional coordinate, solved in closed scalar-Gaussian form via the conditional EPI under a degradedness ordering — containing the scalar single-consumer corner. The only vector-Gaussian equivocation result found (Ekrem–Ulukus [9]) is tight only with the rate constraint removed. The only exact two-distortion-plus-equivocation region found (Tandon–Sankar–Poor) is discrete-alphabet, and it privatizes S itself.

Common reconstruction (Question 1's endpoint). Steinberg [13] and Lapidoth–Malär–Wigger [14] solve the *scalar* Gaussian CR function (the latter for arbitrary scalar jointly Gaussian S via their ξ-reduction); the entire CR follow-up line (Heegard–Berger-with-CR, cascade, successive refinement, interactive, action-dependent) contains no vector Gaussian case. The general-coupling vector machinery that exists (Stylianou–Gkagkos–Charalambous's indirect/WZ RDFs with side information) derives its water-filling precisely from the estimator's *use* of S — the step the CR Markov constraint forbids.

Summary of the open set. Under the recorded queries: no prior treats the pair $(I(X;\widehat{X}), I(X;\widehat{X} \mid S))$ as a two-coordinate region beyond Paper V's one-constraint solution; no exact Gaussian region combines two quadratic distortion constraints with a conditional-information coordinate; no non-degraded vector-Gaussian converse exists for any member of the family; no vector-Gaussian CR function is in print. Residual risks: IEEE-only venues without arXiv copies; the fast-moving RDC line [10] producing a vector-Gaussian paper; the maximal-correlation area received the weakest coverage.

7 The $m = 2$ region, settled as a matrix program

Two rank-one consumers ($Y_A = u^{\mathsf T}X$, $Y_B = v^{\mathsf T}X$, normalized unit variances, read correlation ρ_{AB}), scalar jointly Gaussian reset context $S = V + U$ as before, correlations (ρ_{AV}, ρ_{BV}); WLOG scalar reproductions $(\widehat{Y}_A, \widehat{Y}_B)$ (per-coordinate data processing, as in Section 4).

Theorem 4 ((m+1)-sufficiency)**.** *For m rank-one consumers, it is without loss of optimality to let the joint channel depend on X only through $T = (Y_1, \ldots, Y_m, V)$, and then both coordinates equal their T-versions. The proof of Theorem 1 applies verbatim: the resampled channel preserves the joint law of $(T, \hat{\mathbf{Y}}, S)$ because $S = V + U$ with U jointly independent of (X, all channel randomness), and distortions are functionals of $(Y_i, \widehat{Y}_i)$ pairs. For $m = 2$ the ambient problem, in any dimension d, is exhausted by $(\rho_{AB}, \rho_{AV}, \rho_{BV}, \tau^2)$.*

Theorem 5 (The exact $m = 2$ region)**.** *Assume $\Sigma_T \succ 0$ (degenerate instances reduce: if V is Y_A-measurable, drop V and run the identities on the reduced pair with $\Sigma_{T'|S}$ in place of $\Sigma_{T|S}$). Let Σ_T be the 3×3 correlation matrix of T and $\Sigma_{T|S} = \Sigma_T - \sigma_{TS}\sigma_{TS}^{\mathsf T}/(1 + \tau^2)$, $\sigma_{TS} = \Sigma_T e_V$. Then*

$$\mathcal{RW}(D_A, D_B \mid S) = \mathrm{cl} \bigcup_{\substack{A \in \mathbb{R}^{2\times 3},\ \Sigma_N \succeq 0 \\ (e_i - A_{i\cdot})^{\mathsf T}\Sigma_T(e_i - A_{i\cdot}) + (\Sigma_N)_{ii} \le D_i}} \Big\{(R, L) :\ R \ge \tfrac{1}{2}\log_2 \tfrac{\det(A\Sigma_T A^{\mathsf T} + \Sigma_N)}{\det \Sigma_N},\ L \ge \tfrac{1}{2}\log_2 \tfrac{\det(A\Sigma_{T|S} A^{\mathsf T} + \Sigma}{\det \Sigma_N}$$

jointly Gaussian channels $\hat{\mathbf{Y}} = AT + N$, $N \sim \mathcal{N}(0, \Sigma_N)$, exhaust the region, whose frontier is a nine-parameter matrix program — the work-coordinate analog of the simultaneous-service max-det program of [2].

Proof. Converse. By Theorem 4 restrict to T-channels. For any admissible channel let $C = \mathrm{Cov}(T, \hat{\mathbf{Y}})$ (3×2), $\Sigma_{\hat{Y}} = \mathrm{Cov}(\hat{\mathbf{Y}})$, and set $A := C^{\mathsf T}\Sigma_T^{-1}$, $\Sigma_N := \Sigma_{\hat{Y}} - A\Sigma_T A^{\mathsf T}$ (the Schur complement; joint PSD $\Leftrightarrow$ $\Sigma_N \succeq 0$). The Markov chain and joint independence of U force $\mathrm{Cov}(S, \hat{\mathbf{Y}}) = \mathrm{Cov}(V, \hat{\mathbf{Y}})$ (the V-row of C). Max-entropy bounds on these shared moments give $R \ge \frac{1}{2}\log_2(\det \Sigma_T / \det \Sigma_{e0})$ and $L \ge \frac{1}{2}\log_2(\det \Sigma_{T|S} / \det \Sigma_{e1})$, and the two determinant identities generalize by the same Schur computation: $\det \Sigma_T / \det \Sigma_{e0} = \det(A\Sigma_T A^{\mathsf T} + \Sigma_N)/\det \Sigma_N$ and likewise conditionally (the S-column of the moment matrix is the V-column plus $\tau^2 e_S$). The distortion constraints are exactly the displayed moment constraints. *Achievability.* The Gaussian channel with parameters (A, Σ_N) realizes those moments and attains both bounds with equality. *Numerics (pre-assertion, seed 20260807):* the $\alpha = 1$ projection reproduces the on-regime Xiao–Luo value $\frac{1}{2}\log_2\big((1 - \rho_{AB}^2)/(D_A D_B)\big)$ to 10^{-5}; the tax-note floors hold at every computed point; a coarse unrestricted conditional-BA with two-dimensional reproduction never materially beats the program (grid bias disclosed in the NOVELTY record). $\square$

Remark 5 (No closed form: Conjecture 2 withdrawn as stated)**.** *The $m = 1$ collapse to a scalar quadratic does not persist: the frontier program couples three mutually non-commuting quadratic forms (Σ_T, $\Sigma_{T|S}$, and the constraint directions), and no water-filling closed form emerged — precisely the outcome Conjecture 2's caution anticipated. We therefore* withdraw Conjecture 2 as stated *and replace it with Theorem 5's program, which is the $m = 2$ endpoint of this line: exact, finite-dimensional, solver-computable. Alignment degeneracies still collapse it (next corollary), and Conjecture 1 is settled affirmatively for $m = 2$ rank-one by Theorem 5 itself.*

Corollary 3 (The GO-10 worked instance, exactly). *For orthogonal reads ($\rho_{AB} = 0$) with S aligned to consumer A ($V = Y_A$), the region is the single-corner quadrant at $\big(R_A + R_B,\ L_A + L_B\big)$ with $R_A = R_B = \frac{1}{2}\log_2(1/D)$, $L_B = R_B$, and L_A Steinberg's corner at $\rho_s^2 = 1/(1+\tau^2)$; consequently the complementarity-tax gap is exactly*

$$\mathrm{CT}_R - \mathrm{CT}_W = \tfrac{1}{2}\log_2 \frac{1}{s^2 + (1-s^2)D}, \qquad s^2 = \frac{\tau^2}{1+\tau^2},$$

the formula of [3, Sec. 5], now derived from the exact region rather than the per-coordinate reduction (numerically: all four quantities to 4 decimals at $D \in \{0.25, 0.1\}$, $\tau^2 = 0.25$, and at $\tau^2 = 0.6$, $D = 0.15$, with the $\alpha = 0$ and $\alpha = 1$ optima coinciding — corner degeneracy — as alignment requires).

Proof. This instance is Σ_T-degenerate ($V = Y_A$); apply Theorem 5's reduced form on $T' = (Y_A, Y_B)$, $\Sigma_{T'} = I_2$, $\Sigma_{T'|S} = \mathrm{diag}(s^2, 1)$. *Achievability*: independent per-consumer channels. *Super-additivity converse*: $(Y_A, S) \perp Y_B$ (since $S = Y_A + U$, $Y_B \perp Y_A$, U jointly independent), so for any admissible channel $I(T'; \hat{\mathbf{Y}} \mid S) = I(Y_A; \hat{\mathbf{Y}} \mid S) + I(Y_B; \hat{\mathbf{Y}} \mid Y_A, S) \geq I(Y_A; \hat{\mathbf{Y}} \mid S) + I(Y_B; \hat{\mathbf{Y}} \mid S)$, and each induced marginal channel *is* Markov ($\hat{\mathbf{Y}} = f(Y_A, W)$ with $W = (Y_B, \text{noise}) \perp (Y_A, S)$, and symmetrically), so Steinberg's scalar converse prices the A-term and the classical converse the B-term; the same decomposition without S handles R. The Markov step holds *only* by the instance's double orthogonality ($\rho_{AB} = \rho_{BV} = 0$) — it is exactly the coupling that Proposition 1 shows can fail in general. □

Theorem 6 (Work-floor exactness: Conjecture 3 settled, with a correction). *Write Σ_Y for the read-pair covariance, $\beta = \mathrm{Cov}(Y, V) \in \mathbb{R}^2$, $\Sigma_{Y|V} = \Sigma_Y - \beta\beta^\mathsf{T}$, $\Sigma_{Y|S} = \Sigma_Y - \beta\beta^\mathsf{T}/(1+\tau^2)$, $\Delta =$* $\mathrm{diag}(D_A, D_B) \succ 0$. *The work floor $\frac{1}{2}\log_2\big(\det\Sigma_{Y|S}/\det\Delta\big)$ is attained by the m=2 program iff*

$$\text{(a) } \tau^2 = 0 \text{ and } \Delta \preceq \Sigma_{Y|V}, \qquad \text{or} \qquad \text{(b) } \beta = 0 \text{ and } \Delta \preceq \Sigma_Y.$$

In every other case the floor is strict — both on the misaligned branch ($\tau^2 > 0$ and $\beta \neq 0$, at every positive (D_A, D_B)) and outside the semidefinite conditions of (a)/(b) — with, on the misaligned branch for $\Delta \prec \Sigma_{Y|V}$, the deficit sandwich

$$0 \ <\ L_{\min} - \text{floor} \ \leq\ \tfrac{1}{2}\log_2 \frac{\det\big(I - \Sigma_{Y|S}^{-1}\Delta\big)}{\det\big(I - \Sigma_{Y|V}^{-1}\Delta\big)} \ =\ \tfrac{1}{2}\,\mathrm{tr}\big[(\Sigma_{Y|V}^{-1} - \Sigma_{Y|S}^{-1})\Delta\big]\log_2 e + O(\|\Delta\|^2) \ \longrightarrow\ 0 \ \ (\Delta \to 0).$$

The correction: *Conjecture 3 guessed the condition $\Delta \preceq \Sigma_{Y|S}$; the attainable branch in fact requires the* encoder-accessible *conditional $\Sigma_{Y|V}$ — the same S-vs-V substitution that Proposition 1 and Theorem 2 forced everywhere else in this program.*

Proof. Necessity. Suppose the floor is attained. Equality in the floor's chain forces, at the moment level (max-entropy tightness makes the joint WLOG Gaussian): $E_S := \mathbb{E}[\mathrm{Cov}(Y \mid \hat{\mathbf{Y}}, S)] = \Delta$ with per-consumer equality, so $Z := Y - \hat{\mathbf{Y}}$ has $\mathrm{Cov}\, Z = \Delta$, $\mathrm{Cov}(Z, \hat{\mathbf{Y}}) = 0$, $\mathrm{Cov}(Z, S) = 0$; the latter with the Markov moment identity gives $\mathrm{Cov}(Z, V) = 0$, hence $\mathrm{Cov}(\hat{\mathbf{Y}}, V) = \beta$. Since $E_S \preceq \Sigma_{Y|S}$ always, $\Delta \preceq \Sigma_{Y|S}$ is necessary; at $\tau^2 = 0$ this is $\Delta \preceq \Sigma_{Y|V}$. Tightness of $I(T; \hat{\mathbf{Y}} \mid S) = I(Y; \hat{\mathbf{Y}} \mid S)$ additionally requires $\mathrm{Cov}(V, \hat{\mathbf{Y}} \mid Y, S) = 0$; with the pinned moments this evaluates to $[\Delta\ \ 0]\, M^{-1}\big[\beta;\ \mathrm{Cov}(S, V)\big] = 0$, $M = \mathrm{Cov}(Y, S)$, which (for $\Delta \succ 0$) holds iff $\beta = 0$ or

$\operatorname{Cov}(S,V) = \operatorname{Var} S$, i.e. $\tau^2 = 0$. *Sufficiency (a).* At $\tau^2 = 0$ take $Z \sim \mathcal{N}(0,\Delta)$ jointly Gaussian with $\operatorname{Cov}(Z,Y) = \Delta$, $\operatorname{Cov}(Z,V) = 0$ — existent iff $\Delta - \Delta\Sigma_{Y|V}^{-1}\Delta \succeq 0 \Leftrightarrow \Delta \preceq \Sigma_{Y|V}$ (Schur) — and set $\hat{\mathbf{Y}} = Y - Z$: then $\operatorname{Cov}(\hat{\mathbf{Y}} \mid S) = \Sigma_{Y|S} - \Delta$ and $\operatorname{Cov}(\hat{\mathbf{Y}} \mid T, S) = \Delta - \Delta\Sigma_{Y|V}^{-1}\Delta$, and at $\tau^2 = 0$ ($\Sigma_{Y|S} = \Sigma_{Y|V} =: \Sigma_c$)

$$L = \tfrac{1}{2}\log_2 \frac{\det(\Sigma_c - \Delta)}{\det\Delta\ \det(I - \Sigma_c^{-1}\Delta)} = \tfrac{1}{2}\log_2 \frac{\det\Sigma_c}{\det\Delta} = \text{floor}.$$

Sufficiency (b). At $\beta = 0$, $\Sigma_{Y|S} = \Sigma_Y$ and a plane-only channel has $\hat{\mathbf{Y}} \perp S$, so $L = I(Y;\hat{\mathbf{Y}})$, attained at the floor iff $\Delta \preceq \Sigma_Y$ (the rate-side exactness of [3]); necessity of $\Delta \preceq \Sigma_Y$ is the $E_S \preceq \Sigma_{Y|S} = \Sigma_Y$ bound. *Sandwich.* The same Z-construction at $\tau^2 > 0$ is admissible whenever $\Delta \prec \Sigma_{Y|V}$ and yields $L \le \tfrac{1}{2}\log_2[\det(\Sigma_{Y|S} - \Delta)/\det(\Delta - \Delta\Sigma_{Y|V}^{-1}\Delta)] = \text{floor} + \tfrac{1}{2}\log_2[\det(I - \Sigma_{Y|S}^{-1}\Delta)/\det(I - \Sigma_{Y|V}^{-1}\Delta)]$; the ratio is ≥ 1 because $\Sigma_{Y|S} \succeq \Sigma_{Y|V}$ (congruence by $\Delta^{1/2}$ keeps both factors in the PSD order where det is monotone), and $\to 1$ as $\Delta \to 0$. Strictness (> 0) is the necessity direction, using that $L_{\min}$ is *attained* (compact feasible (A, Σ_N) set, lower semicontinuity); at the non-strict boundary of $\Delta \preceq \Sigma_{Y|V}$ in case (a), where the displayed ratio is $0/0$, attainment follows by $\Delta_t \uparrow \Delta$ feasibility and floor continuity. *(m=1 corollary.)* For one consumer, on the active-floor regime $D \le \operatorname{Var}(Y \mid V)$, the floor $\tfrac{1}{2}\log_2(\operatorname{Var}(Y \mid S)/D)$ is attained iff $\rho\tau = 0$: substituting $g_f = (s-\rho^2)/(Ds)$ into Theorem 2's quadratic gives $P(g_f) = -\rho^2\tau^2/s$, which is < 0 (hence $g^\star > g_f$ strictly) exactly when $\rho\tau \ne 0$. (The regime qualifier is the m=1 shadow of condition (a): for $\tau = 0$, $D > 1-\rho^2$ the clamped floor and $L = 0$ meet trivially while $g_f < 1$ becomes the quadratic's *smaller* root.) □

Remark 6 (Conjecture 3: former numerical status)**.** *(Superseded by Theorem 6; retained as the numerical record that motivated it.) At the misaligned test instance* $(\rho_{AB}, \rho_{AV}, \rho_{BV}, \tau^2) = (0.3, 0.7, 0.2, 0.4)$ *the work floor is strictly loose at every* $D > 0$ *(gap* $0.0664 \to 0.0293 \to 0.0090 \to 0.0035$ *bits as* $D = 0.3 \to 0.15 \to 0.05 \to 0.02$*), decreasing toward* 0 *as* $D \to 0$ *— now explained exactly by Theorem 6 (*$\beta \ne 0$, $\tau^2 > 0$*), with the measured gaps inside the deficit sandwich at every probe. The general-instance frontier is nondegenerate (*R *spans* 2.2539–2.2942*,* L 1.9838–2.0175 *bits at* $D_A = D_B = 0.2$*): the misalignment tradeoff of Corollary 1 persists at* $m = 2$.

8 Vector side information and higher-rank reads

Both remaining structural rungs reduce to the machinery already built, one of them completely.

Theorem 7 (Vector S, $m = 1$ rank-one: solved)**.** *Let* $S = \Gamma^\mathsf{T} X + U \in \mathbb{R}^r$*,* $U \sim \mathcal{N}(0, \Sigma_U)$*,* $\Sigma_U \succ 0$*, with* X, U *and all channel randomness mutually independent; assume* $\Sigma_T \succ 0$ *(degenerate instances reduce by dropping dependent context components, as in Theorem 5); reproductions are scalar WLOG by the data-processing reduction of Section 4, which is generic in S. (i)* (1+r)-sufficiency: *the channel may depend on* X *only through* $T = (Y, \Gamma^\mathsf{T} X)$ *(the resampling proof of Theorem 1 applies verbatim with the vector kernel* $p(s \mid t, \hat{y}) = p_U(s - v)$*). (ii)* Whitened form: *congruence- diagonalize the pair —* $\Sigma_T \to I$*,* $\Sigma_{T|S} \to \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_{1+r})$*,* $0 \prec \Lambda \preceq I$ *— so the read is a unit vector* y_0*, the record* $\widehat{Y} = u^\mathsf{T} Z + N(0, n)$*, and*

$$R = \tfrac{1}{2}\log_2 \frac{|u|^2 + n}{n}, \qquad L = \tfrac{1}{2}\log_2 \frac{u^\mathsf{T}\Lambda u + n}{n}, \qquad |y_0 - u|^2 + n \le D.$$

(iii) The frontier*, traced by* $\alpha \in [0,1]$*, is the generalized two-water-level system: with* $\gamma_0 = n/(|u|^2{+}n)$*,* $\gamma_1 = n/(u^\mathsf{T}\Lambda u{+}n)$*,*

$$u \;=\; \big(1-\alpha\gamma_0-(1-\alpha)\gamma_1\big)\Big[\big(1-(1-\alpha)\gamma_1\big)I+(1-\alpha)\gamma_1\Lambda\Big]^{-1}y_0, \qquad n = D-|y_0-u|^2, \quad (3)$$

per-mode explicit given the two scalars; Theorem 3 is the $r=1$ *case, and the* $\alpha=0$ *endpoint gives the vector-S CR function as a one-dimensional root problem.* Converse: *the determinant identities are dimension-free (the same Schur double-count as Theorems 2 and 5, with the Markov moment identity* $\mathbb{E}[S\widehat{Y}] = \mathbb{E}[V\widehat{Y}] = \Gamma^\mathsf{T}\mathbb{E}[X\widehat{Y}]$ *— the* V*-block of* $\mathbb{E}[T\widehat{Y}]$ *— holding componentwise), so the weighted moment program is again identical to the whitened channel program; coercivity and the linear-in-u first-order system pin the optimum as before.*

Proof sketch and verification status. (i) is Theorem 1's proof with $v \in \mathbb{R}^r$. (ii) is the standard simultaneous congruence ($\Sigma_{T|S} \preceq \Sigma_T$; $\operatorname{Var} Y = 1$ gives $|y_0| = 1$; the distortion form is exact because Z is white). (iii): substituting $n = D - h$ and differentiating, the Q_0-gradient term collapses ($u + (y_0 - u) = y_0$), yielding $\alpha\gamma_0 y_0 + (1-\alpha)\gamma_1\big(y_0 + (\Lambda - I)u\big) = (y_0 - u)$, which rearranges to (3). Pre-assertion numerics (seed 20260904): (3) holds at the multi-start optimum at $r \in \{1,2,3\}$ with fully general couplings, all tested α; $r{=}1$ reproduces Theorem 3's values to 2×10^{-3} bits; the moment program agrees to 5×10^{-7}; unrestricted conditional-BA never beats the system (window). Fresh-context pass: see the NOVELTY addendum. □

Theorem 8 (Higher-rank reads: the matrix program, and when it decomposes)**.** *For a* k*-dimensional read* $Y = W^\mathsf{T}X$ *with trace distortion* $\mathbb{E}|Y-\hat{\mathbf{Y}}|^2 \le D$ *and (scalar or vector) Gaussian context,* $(k{+}r)$*-sufficiency holds, and the region equals the* $k \times (k{+}r)$*-parameter matrix program of Theorem 5's form (the identities are dimension-free). If additionally* Σ_T *and* $\Sigma_{T|S}$ *are simultaneously block-diagonal, with each block pairing one read mode with its own context components (in a read basis with* $\Sigma_Y = I$*), the program* decomposes exactly*: the frontier is the per-mode family of Theorem 7 systems coupled only through the distortion budget — a water-filling over modes. For misaligned couplings the joint program is strictly better than every forced per-mode decomposition (numerically: gap* 0.0057 *bits at the test instance, seed 20260904; the aligned equality is exact, matched by the solver to* 3×10^{-4}*).*

Proof of the decomposition. With independent blocks (T_i, S_i): $h(T \mid S) = \sum_i h(T_i \mid S_i)$ and $h(T \mid \hat{\mathbf{Y}}, S) \le \sum_i h(T_i \mid \widehat{Y}_i, S_i)$ (subadditivity; conditioning reduces entropy), so $I(T;\hat{\mathbf{Y}} \mid S) \ge \sum_i I(T_i;\widehat{Y}_i \mid S_i)$, and identically for the rate coordinate. Each pair $(T_i, \widehat{Y}_i)$ inherits the per-mode Markov moment identity ($U \perp$ everything), so Theorem 7's converse prices mode i at its own distortion $D_i = \mathbb{E}(Y_i - \widehat{Y}_i)^2$ with $\sum_i D_i \le D$; each per-mode value is convex in D_i, so the allocation is a water-filling, and independent per-mode optimal channels achieve it. □

Proposition 2 (Interior-α uniqueness: the S2 debt discharged)**.** *For* $0 \prec \Lambda \preceq I$*,* $|y_0| = 1$*,* $D \in (0,1)$*, and every* $\alpha \in (0,1]$*, the weighted program has a* unique *minimizer, and every solution of* (3) *with* $n > 0$ *is that minimizer. Consequently Theorem 3's trace of the Pareto boundary is unconditionally complete, and its "unique on every tested interior* α*" hedge is retired (*$\alpha = 0$ *remains covered by Theorem 2's* $P(1) < 0$*).*

Proof (matrix-convexity; completed by the fresh-context pass). Work in moment coordinates $c = \mathbb{E}[Z\widehat{Y}]$, $v = \operatorname{Var}\widehat{Y}$ on $\{v > |c|^2,\ 1 - 2y_0^\mathsf{T}c + v \le D\}$, where the program equals min $\alpha B_0 + (1-\alpha)B_1$

with $B_0 = -\frac{1}{2}\log\det(I - cc^\mathsf{T}/v)$ and $B_1 = \frac{1}{2}\log\det\Lambda - \frac{1}{2}\log\det(\Lambda - (\Lambda c)(\Lambda c)^\mathsf{T}/s)$, $s = v - c^\mathsf{T}(I-\Lambda)c$. *Key fact:* if $a(z)$ is affine and $s(z)$ concave positive, then $z \mapsto a(z)a(z)^\mathsf{T}/s(z)$ is matrix-convex — average the two Schur certificates $\left[\begin{smallmatrix} W & a \\ a^\mathsf{T} & s \end{smallmatrix}\right] \succeq 0$ and use concavity of s to raise the corner. Applied to $W_0 = cc^\mathsf{T}/v$ and $W_1 = (\Lambda c)(\Lambda c)^\mathsf{T}/s$, the error matrices $I - W_0$ and $\Lambda - W_1$ are matrix-concave, and $-\log\det$ is convex antitone, so B is *convex.* On the feasible set $D < 1$ forces $y_0^\mathsf{T} c \geq (1-D)/2 > 0$ (so $c \neq 0$, coercivity as $n \downarrow 0$), and B strictly decreases in v at fixed c, so minimizers lie on the active slice $v = D - 1 + 2y_0^\mathsf{T} c$; there B is convex in $u = c$, its stationary points are exactly (3), and every stationary point is global. Strictness: on a segment of minimizers each B_i is affine, which forces (concavity–equality plus $\mathrm{tr}[(\bar\Sigma^{-1}\bar\Sigma')^2] = 0$) W_0 constant, hence $c(t) = \tau(t)e$ with τ affine and $v = \tau^2/(e^\mathsf{T} W_0 e)$ affine — so τ is constant. Unique. [Verifier's convexity spot-check: 400 finite-difference Hessian probes across $r \leq 4$, $\lambda_{\min}$ down to 0.01, D up to 0.98: minimum eigenvalue +0.36; fixed-point hunts: one root in all 2160 runs.] $\square$

9 The $m = 2$ frontier system: matrix water levels

Theorem 5 characterized the m=2 region as a nine-parameter program; the m=1 closure now begs the structural question of whether the two-water-level form survives a second consumer. It does — the water levels become 2×2 matrices.

Theorem 9 (The $m = 2$ frontier system)**.** *Whiten as in Theorem 7 ($\Sigma_T \to I$, $\Sigma_{T|S} \to \Lambda$, reads the rows of $\widetilde{Y} = [y_A^\mathsf{T}; y_B^\mathsf{T}]$, record $\hat{\mathbf{Y}} = AZ + N$, $N \sim \mathcal{N}(0, \Sigma_N)$, $A \in \mathbb{R}^{2\times(2+r)}$). Write $M_0 = AA^\mathsf{T} + \Sigma_N$, $M_1 = A\Lambda A^\mathsf{T} + \Sigma_N$ (both 2×2), and let $\mu = (\mu_A, \mu_B) \succeq 0$ be the distortion multipliers. Assume an optimum is attained with $\Sigma_N \succ 0$; the constraints may be taken active without loss (the objective is non-increasing in each noise diagonal, since $\Sigma_N \preceq M_0, M_1$ gives $[\Sigma_N^{-1}]_{ii} \geq [M_k^{-1}]_{ii}$), though strict complementarity $\mu_i > 0$ is* not *assumed. Every Pareto-optimal channel of the weighted program (weight $w \in [0,1]$ on the rate coordinate) satisfies*

$$\Sigma_N^{-1} \;=\; wM_0^{-1} + (1-w)M_1^{-1} + \mathrm{diag}(\mu) \tag{4}$$

— including its off-diagonal entry, a parameter-free structural prediction — and, per mode j*,*

$$a_j \;=\; \left[\Sigma_N^{-1} + (1-w)(\lambda_j - 1)M_1^{-1}\right]^{-1}\mathrm{diag}(\mu)\,\tilde{y}_j, \tag{5}$$

where a_j, $\tilde{y}_j$ are the j-th columns of A, $\widetilde{Y}$; the bracket in (5) *is automatically invertible, since by* (4) *it equals $wM_0^{-1} + (1-w)\lambda_j M_1^{-1} + \mathrm{diag}(\mu) \succ 0$ for every $w \in [0,1]$, $\lambda_j > 0$, $\mu \succeq 0$. KKT necessity requires no constraint-qualification hypothesis: ∇c_A and ∇c_B have disjoint supports (row* 1 *of A and $(\Sigma_N)_{11}$ versus row* 2 *and $(\Sigma_N)_{22}$), so LICQ holds identically. The system closes in eleven scalars ($M_0, M_1, \Sigma_N \in \mathbb{S}^2$, $\mu \in \mathbb{R}^2$, with their three defining equations,* (4)*, and the two active distortion constraints) — a bookkeeping count, not a well-posedness claim: when some $\mu_i = 0$ the solution set can be non-isolated (verified numerically: a slack-constraint instance exhibits an exactly flat ray of minimizers). The m=1 system of Theorems 3/7 is its scalar shadow (substituting scalars reproduces* (3) *exactly). At $w = 1$ the record is confined to the read span, $A = \Sigma_N\,\mathrm{diag}(\mu)\widetilde{Y}$, recovering the rate-only (Xiao–Luo) structure; on the Xiao–Luo regime (where the two-consumer rate bound $\frac{1}{2}\log_2[\det\Sigma_{\mathbf{Y}}/(D_A D_B)]$ is tight) the w=1 error covariance is forced to $EE^\mathsf{T} + \Sigma_N = \mathrm{diag}(D_A, D_B)$ ($E = \widetilde{Y} - A$; the Hadamard/backward-channel equality condition in the Xiao–Luo converse) and $\mu_i = 1/D_i$* without any symmetry assumption *(envelope theorem on $J^* = \ln[\det\Sigma_{\mathbf{Y}}/(D_A D_B)]$).*

Consequently: the Pareto frontier is contained in *the solution set of* (4)–(5)*, and every solution of the system is an achievable point; solutions must be compared by objective value. [Uniqueness RESOLVED subsequent to this statement: the m-record moment-convexity lemma (GO-13 manuscript, netted GO-P-2026-074) proves the weighted program convex in moment coordinates for every m — the block lift of Proposition 2's scalar argument; optimal invariants unconditionally unique, (C, V) unique whenever the optimal C has full column rank, the failure mode being exactly a record priced out by the water-filling.]*

Proof. Matrix differentiation of $J = w \log\det M_0 + (1-w)\log\det M_1 - \log\det \Sigma_N$ with the KKT terms of the two constraints $c_i = |y_i - A_{i\cdot}|^2 + (\Sigma_N)_{ii}$. For the Σ_N-gradient, parameterize Σ_N by its three entries; the symmetric-differentiation factor of 2 on the off-diagonal multiplies every $\log\det$ term identically and cancels, and the constraints touch only the *diagonal* entries with coefficient 1, so no convention mismatch is possible: stationarity gives (4), including its off-diagonal row. The A-gradient gives $wM_0^{-1}A + (1-w)M_1^{-1}A\Lambda = \mathrm{diag}(\mu)(\widetilde{Y} - A)$. Adding $\mathrm{diag}(\mu)A$ to both sides and substituting $\mathrm{diag}(\mu)$ from (4) collapses the M_0 terms:

$$(1-w)M_1^{-1}A(\Lambda - I) + \Sigma_N^{-1}A = \mathrm{diag}(\mu)\widetilde{Y},$$

which is (5) column-wise (Λ diagonal). Coercivity and interior Σ_N are as in Theorem 5; any system solution is achievable (it is its own Gaussian channel). [Pre-assertion numerics, seed 20260907: (4)'s off-diagonal and (5) hold at the 80-start program optimum to $\le 2 \times 10^{-7}$ at two scalar-context instances and one vector-context instance (r=2, d_T=4), all three weights; the w=1 read-span anchor to 1.4×10^{-8}. An independent fresh-context verification pass re-derived every identity and confirmed them at Newton-polished KKT points to $\le 3.3 \times 10^{-16}$, including a slack-constraint instance where the system degrades gracefully ($\mu_B \to 0$, identities intact) while exhibiting the non-isolated minimizer ray noted in the statement.] □

Remark 7 (Conjecture 2, partially rehabilitated)**.** *The withdrawal in Remark 5 stands for* scalar *water-filling: no per-mode scalar formula exists. But* (5) *is a* matrix-resolvent water-filling *— the record's mode-j coefficients are a fixed 2×2 resolvent of the water matrices applied to the read column — so the structural content of the conjecture survives with the water levels promoted from scalars to 2×2 matrices, in any context dimension.*

10 The binary pair: exact conditional CR function

The discrete rung closes completely — and, unlike the Gaussian Theorem 2, with an elementary self-contained proof.

Let (Y, V) be a doubly symmetric binary pair (fair marginals, $\Pr[Y \neq V] = p \in (0, \frac{1}{2})$), context V, side information $S = V \oplus W$ with $W \sim \mathrm{Bern}(q)$, $q \in [0, \frac{1}{2}]$, independent. The record is the reproduction $\hat{Y} \in \{0, 1\}$ itself (richer records only cost more: for any stored Z from which the consumer computes $\hat{Y} = f(Z)$, $I(T; Z \mid S) \ge I(T; \hat{Y} \mid S)$ by the data-processing inequality), Hamming distortion $\Pr[\hat{Y} \neq Y] \le D$. Write $x * q = x(1-q) + (1-x)q$ and $\ell(x) = \ln \frac{1-x}{x}$.

Theorem 10 (Binary conditional CR function)**.** *For $D \in (0, \frac{1}{2})$, the conditional coordinate $L(D) = \min I(Y, V; \hat{Y} \mid S)$ over channels $P(\hat{Y} \mid Y, V)$ (Markov $\hat{Y} - (Y, V) - S$) with* $\Pr[\hat{Y} \neq Y] \le D$ *is* *attained in the two-parameter symmetric family* $\hat{Y} = Y \oplus E$, $\Pr[E = 1 \mid Y \oplus V = j] = d_j$, *and equals*

$$L(D) \;=\; h_2(u) - (1-p)\,h_2(d_0) - p\,h_2(d_1), \qquad u = a * q, \quad a = (1-p)d_0 + p(1-d_1),$$

where — for $q \in (0, \frac{1}{2}]$, *any* $D \in (0, \frac{1}{2})$*; and for* $q = 0$, $D \le p$ *—* (d_0, d_1) *solves the active constraint* $(1-p)d_0 + pd_1 = D$ *together with the* tilt equation

$$\ell(d_0) - \ell(d_1) \;=\; 2(1-2q)\,\ell(u).$$

(In the excluded sliver $q = 0$, $D > p$*, the constraint is inactive and* $L(D) = 0$*, attained non-uniquely by records measurable in* V *alone; the tilt system still has a root there, the degenerate point* $d_1 = 1 - d_0$*, which evaluates to the correct value* 0*. For* $q > 0$ *the constraint is active on all of* $(0, \frac{1}{2})$*:* L *is convex, nonincreasing, positive by the last claim below, and* 0 *at* $D = \frac{1}{2}$*, hence strictly decreasing.) On the same channel the rate coordinate is* $R = 1 - (1-p)h_2(d_0) - p\,h_2(d_1)$ *and the discount is exactly* $R - L = 1 - h_2(u) = I(\hat{Y}; S)$ *(Fact 1 in closed form:* $a = \Pr[\hat{Y} \ne V]$*,* $u = \Pr[\hat{Y} \ne S]$*). Anchors: at* $q = 0$ *the system gives* $a = (p-D)/(1-2D)$ *and* $L = h_2(p) - h_2(D)$ *(Gray's conditional rate–distortion function, for* $D \le p$*); at* $q = \frac{1}{2}$ *it gives* $d_0 = d_1 = D$ *and* $L = 1 - h_2(D)$ *(the marginal rate–distortion function — the context is useless exactly when the side information is). For* $q \in (0, \frac{1}{2}]$ *and* $D < \frac{1}{2}$*,* $L(D) > 0$*.*

Proof. Reduction and convexity. Binary $\hat{Y}$ is definitional (DPI above). With $P(Y, V \mid S = s)$ fixed and the channel shared across s (Markov), $I(Y, V; \hat{Y} \mid S) = \sum_s P(s)\, I_s$ is a nonnegative combination of mutual informations each convex in the channel, hence convex in the channel; the distortion is linear.

Symmetrization. The global bit flip $\sigma : (y, v, s, \hat{y}) \mapsto (1-y, 1-v, 1-s, 1-\hat{y})$ preserves the joint law of (Y, V, S) and the distortion, and $L(C^\sigma) = L(C)$; averaging C with C^σ preserves distortion and, by convexity, does not increase L. So an optimum exists among σ-symmetric channels, and these are exactly the family $\{(d_0, d_1)\}$: σ-symmetry forces $\Pr[\hat{Y} \ne Y \mid y, v]$ to depend only on $y \oplus v$.

Closed form. On the family, $H(\hat{Y} \mid Y, V) = (1-p)h_2(d_0) + p\,h_2(d_1)$, and $\hat{Y} \oplus S = J \oplus E \oplus W$ ($J = Y \oplus V$) with $\Pr[J \oplus E = 1] = a$. The step $H(\hat{Y} \mid S) = h_2(a * q)$ needs $\hat{Y} \oplus S \perp S$, which holds because the joint law of $(\hat{Y}, S)$ is σ-invariant (σ-invariant source, σ-symmetric channel), forcing equal conditional flip probabilities given either value of S; the same symmetry gives $H(\hat{Y}) = 1$, hence R.

Tilt equation. Substituting $d_1 = (D - (1-p)d_0)/p$ makes $a = 2(1-p)d_0 + p - D$ affine in d_0; differentiating L and using $h_2'(x) = \ell(x)$ (nats) gives $(1-p)[2(1-2q)\ell(u) - \ell(d_0) + \ell(d_1)] = 0$. Since L restricted to the affine family slice is convex (restriction of a convex function), interior stationarity is sufficient, not merely necessary.

Anchors and positivity. $q = \frac{1}{2}$: $u = \frac{1}{2}$, the tilt equation forces $d_0 = d_1 = D$. $q = 0$: substitution of Gray's backward channel; equality of values is elementary algebra. Positivity, in two branches over the family (which suffices by attainment): if $a \ne \frac{1}{2}$, then $h_2(u) > h_2(a) \ge (1-p)h_2(d_0) + p\,h_2(1-d_1) = (1-p)h_2(d_0) + p\,h_2(d_1)$ — the strict step because $u = a * q$ is strictly closer to $\frac{1}{2}$ for $q \in (0, \frac{1}{2}]$, the second step by concavity — so $L > 0$; if $a = \frac{1}{2}$, then $u = \frac{1}{2}$, $h_2(u) = 1$, and $L = 1 - (1-p)h_2(d_0) - p\,h_2(d_1) \ge 1 - h_2(D) > 0$ by concavity at the distortion. [Pre-assertion numerics, seed 20260908: unconditional Lagrangian certificates — if $\min_C[L(C) + \beta\,\mathrm{dist}(C)]$ over *all* channels equals the family value $L + \beta D$, every channel with distortion $\le D$ has $L(C) \ge L$, no convexity assumed — hold to $\le 6 \times 10^{-15}$ at four (p, q, D) instances against reproduction alphabets of size 2, 4, and 6; tilt-equation residuals $\leq 10^{-7}$; both anchors to 10^{-9} or better; L monotone in q. An independent fresh-context verification pass re-derived every step, confirmed the global optimum over all binary channels matches family-plus-tilt to $\leq 7 \times 10^{-14}$ at six instances including extreme corners, found no improvement from 4-symbol records ($\leq 3.2 \times 10^{-14}$), and contributed the q=0 regime qualification, the a=$\frac{1}{2}$ positivity branch, and the σ-invariance justification now in the text.] $\square$

Remark 8 (The noisy face of the 062 registration, retro-derived)**.** *The sealed encoder-tilt binary run (GO-P-2026-062; $p = 0.25$, $D = 0.20$) reported a noisy-context discount* 0.3432 *at* $q = 0.1$ *with no committed formula. Theorem 10 now supplies it: the* q=0*-optimal record has* $a = (p-D)/(1-2D) = \frac{1}{12}$*, so under* $q = 0.1$ *the discount is* $1-h_2(\frac{1}{12}*0.1) = 1-h_2(\frac{1}{6}) = 0.3500$*. The* -0.007 *instrument bias matches the same run's* explained *face exactly (measured* 0.5782 *against derived* $1-h_2(\frac{1}{12}) = 0.5862$*, bias* -0.008*, the known decode-threshold interpolation bias): the noisy face is now derived to the same precision as the clean one.*

11 Line of attack and verification obligations

Attack. (1) Question 1's marginalization dichotomy first — it is decidable with scalar tools. (2) Then the endpoint via the scalar corner's moment/LMMSE method lifted to the S-whitened frame, watching the three-matrix non-commutativity of Remark in Conjecture 2. (3) For Conjecture 1, conditional-EPI/extremal-inequality arguments in the Günlü–Sankar style, plus the per-s convexity decomposition (VI-8). (4) Numerical falsification *before* proof: the 044/058 conditional Blahut–Arimoto fixed point brute-forces small discretized instances against any conjectured frontier.

Obligations (house rules). R-IND-5 fresh-context pass on any proof before assertion; C3 harness sealed under PROTOCOL §5.1 (power and pilot fields mandatory; logged pilot; instrument-vs-physics gate separation); the residual-risk sweeps above refreshed at claim time; ledger class no higher than [predicted] until then. GO-11 is *not yet registered*: this document reserves no ID and asserts no claim.

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